

# When Negation is a Geometry Problem in Vision Language Models

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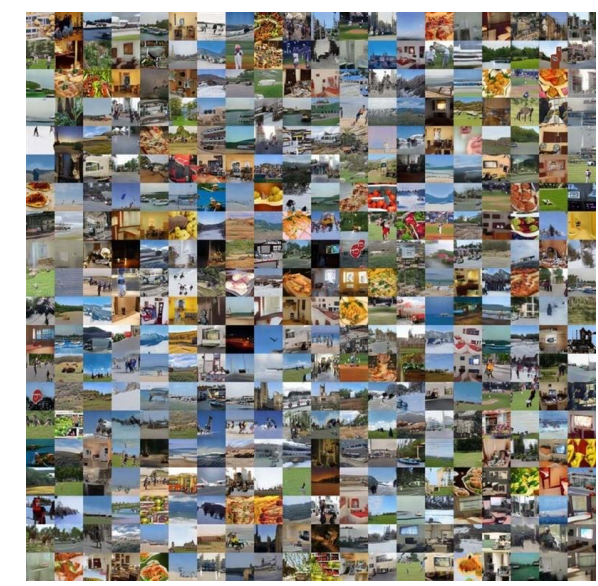
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## Vision Language Models struggle to understand negation

"A shirt with no logo"

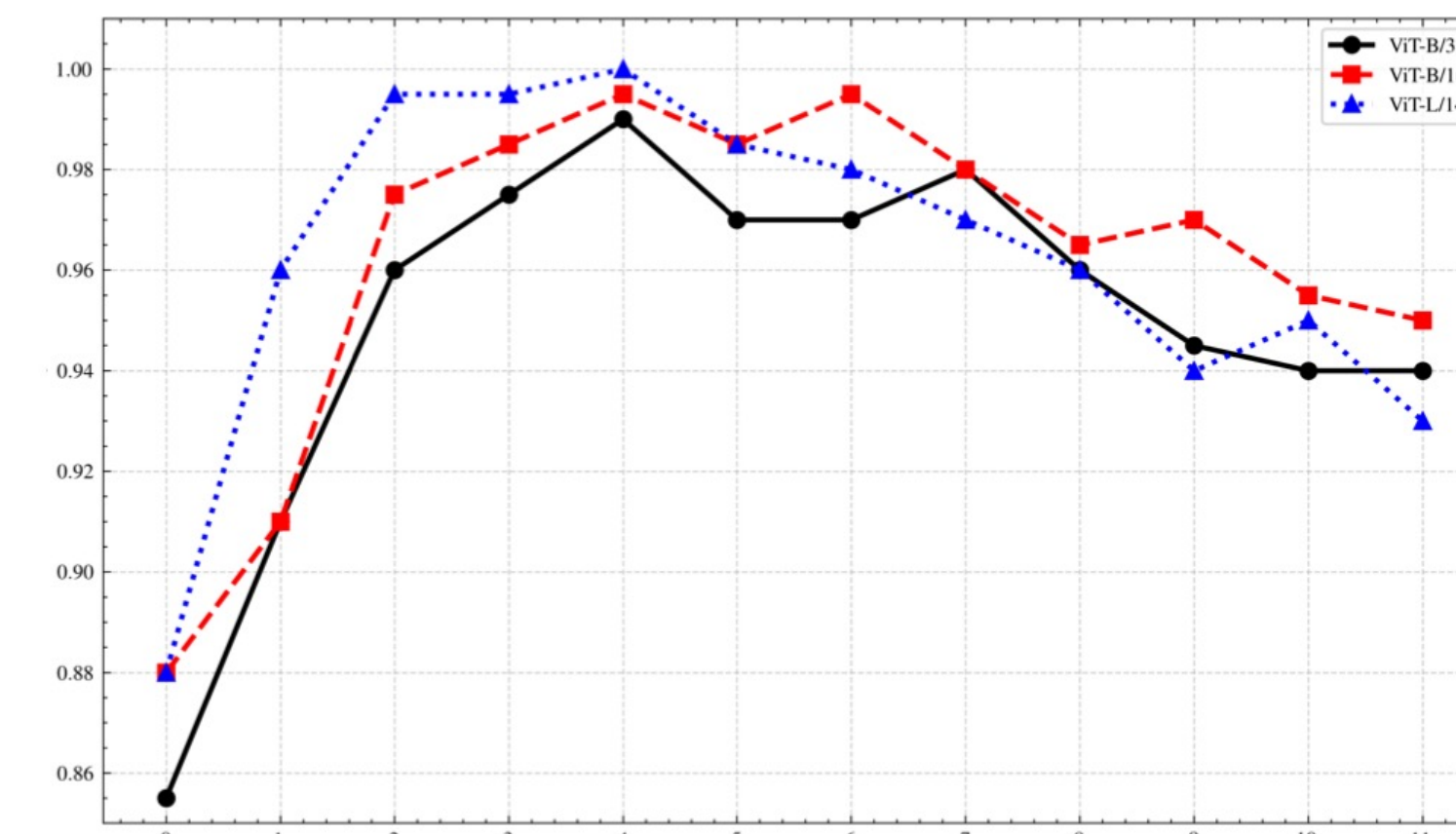


CLIP



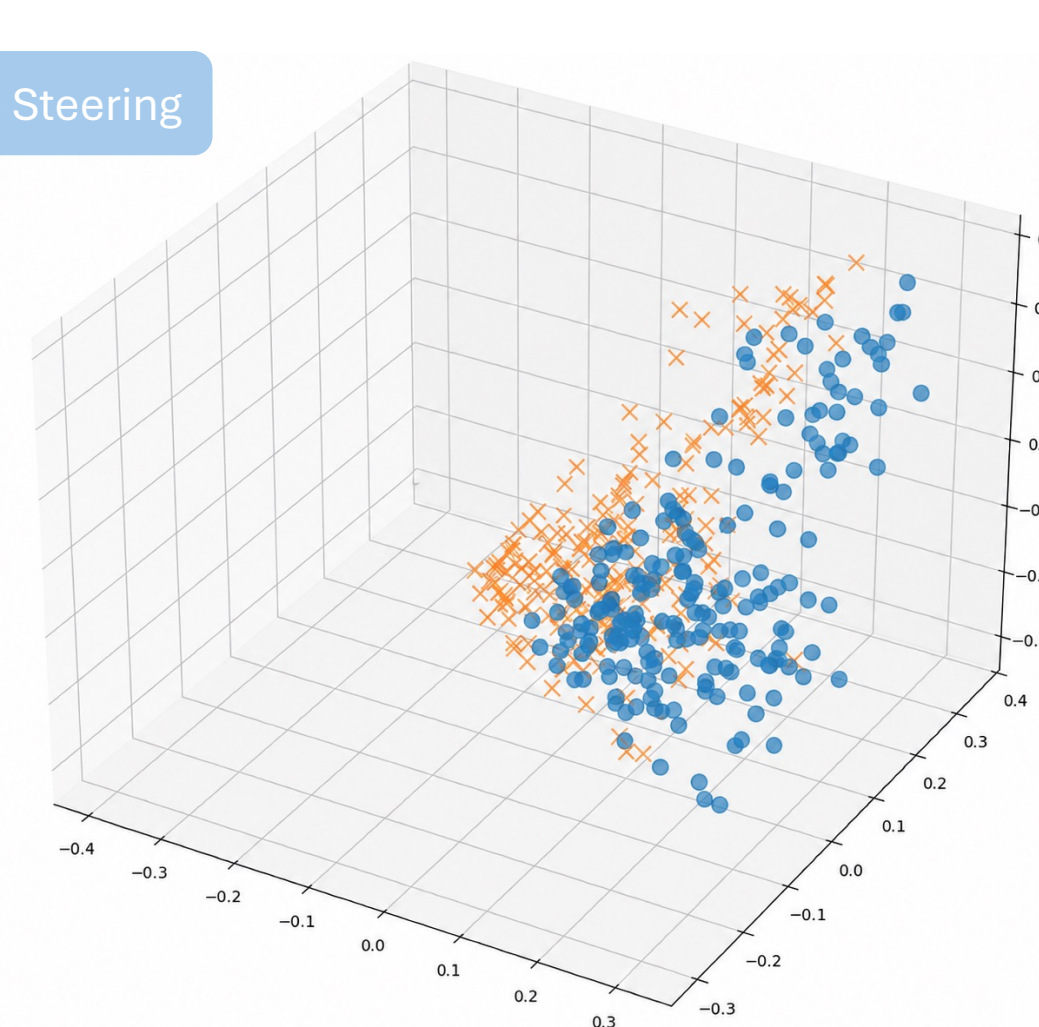
## Negation information exists in the latent space but it's not activated enough

We train a linear probe on every layer to distinguish between a caption and its negation counterpart

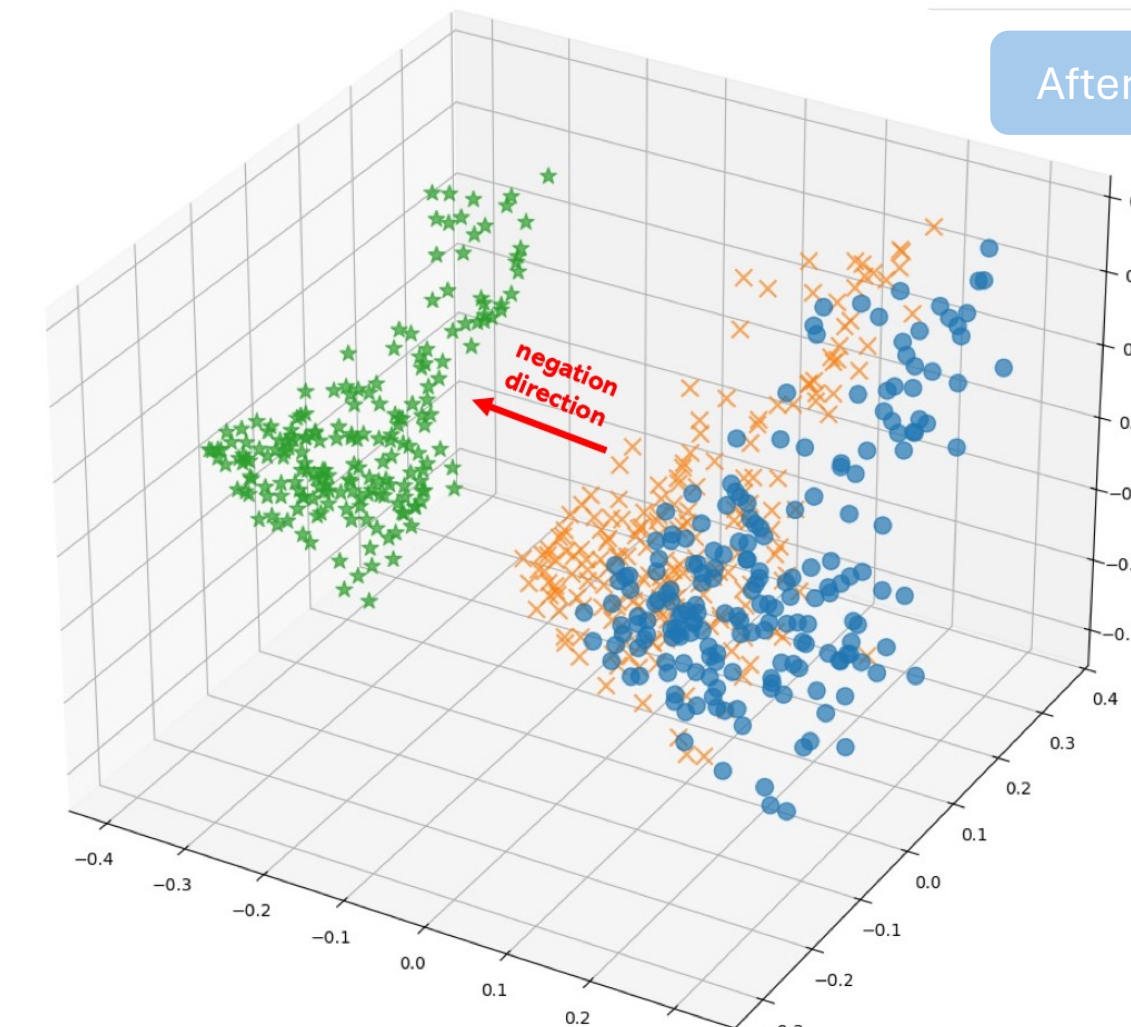


## We can steer towards the negation direction!

Before Steering



After Steering



A kitchen with no fridge



## VLM-as-a-Judge prevents false negatives!

A dog that is not running

Retrieved Images



Semantic Judge

is there a dog in the image?

VLM ✓

Negation Judge

is the dog running?

VLM → No ✓

is there a dog in the image?

VLM ✓

is the dog running?

VLM → Yes ✗

is there a dog in the image?

VLM ✗

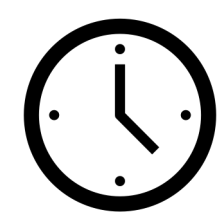
- Annotation free
- Scalable
- Any database
- More fair

| Model          | Qwen3-VL-8B |             |             |                        |             |             | Qwen3-VL-32B |             |             |                        |             |             |
|----------------|-------------|-------------|-------------|------------------------|-------------|-------------|--------------|-------------|-------------|------------------------|-------------|-------------|
|                | Retrieval   |             |             | Retrieval and Negation |             |             | Retrieval    |             |             | Retrieval and Negation |             |             |
|                | Top-1       | Avg.5       | Top-5       | Top-1                  | Avg.5       | Top-5       | Top-1        | Avg.5       | Top-5       | Top-1                  | Avg.5       | Top-5       |
| CLIP [20]      | 46.7        | 30.4        | 75.2        | 39.4                   | 25.1        | 68.4        | 47.2         | 30.6        | 76.3        | 40.3                   | 25.4        | 69.9        |
| ConCLIP [22]   | 46.6        | 30.7        | 76.4        | 40.8                   | 26.1        | 70.7        | 47.2         | 30.7        | 77.7        | 41.6                   | 26.3        | 72.2        |
| NegCLIP [23]   | <b>60.9</b> | <b>37.9</b> | <b>84.3</b> | <b>51.7</b>            | <b>31.5</b> | <b>77.9</b> | <b>62.0</b>  | <b>38.3</b> | <b>85.8</b> | <b>53.1</b>            | <b>32.1</b> | <b>79.8</b> |
| CLIP-CC12M [1] | 49.0        | 31.2        | 76.4        | 42.7                   | 26.7        | 70.6        | 49.9         | 31.4        | 78.0        | 43.8                   | 27.0        | 72.5        |

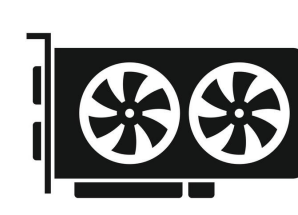
## All current approaches rely on expensive training



Large Synthetic Data



Long Training Time



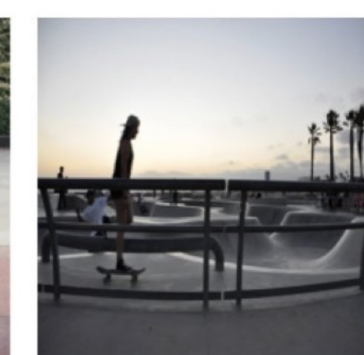
Heavy Computational Resources



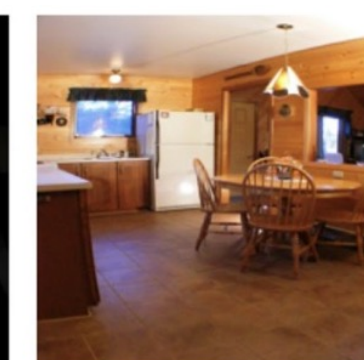
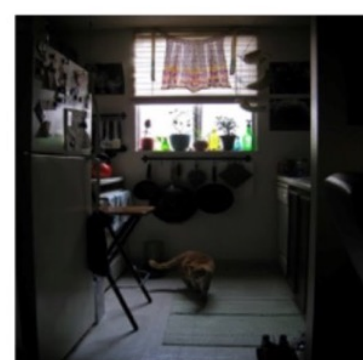
High Cost

## Current evaluation benchmarks contain False Negatives

No handbag is visible in the image, where a man on a skateboard is performing a trick at the skate park.



The kitchen is clean and empty, with no person in sight.



## Stress-Testing on Non Common Objects



An image of a book not made of paper

Positive:



Distractor:



Stress-test your own models!

